TIWARI KISHLAYNATH SHAILENDRAKUMAR

Data Analyst – Convexity Sensitivity AI Agent

Fixed-income portfolio risk analytics with Monte Carlo simulation, machine learning, Power BI and Excel validation.

Project 1B – Data Analyst: Convexity Sensitivity AI Agent

WORKFLOW

One workflow turns bond data into risk decisions

The problem is explaining portfolio sensitivity: what moves value, where exposure concentrates, and how scenario losses should be interpreted.

01

Bond pricing → Duration, convexity and DV01

02

KRD and yield-curve analysis → Monte Carlo P&L

03

ML comparison → AI-assisted interpretation

04

Power BI and Excel validation

PORTFOLIO SNAPSHOT

The portfolio contains 302 bonds with INR 32.06 million of market value

302

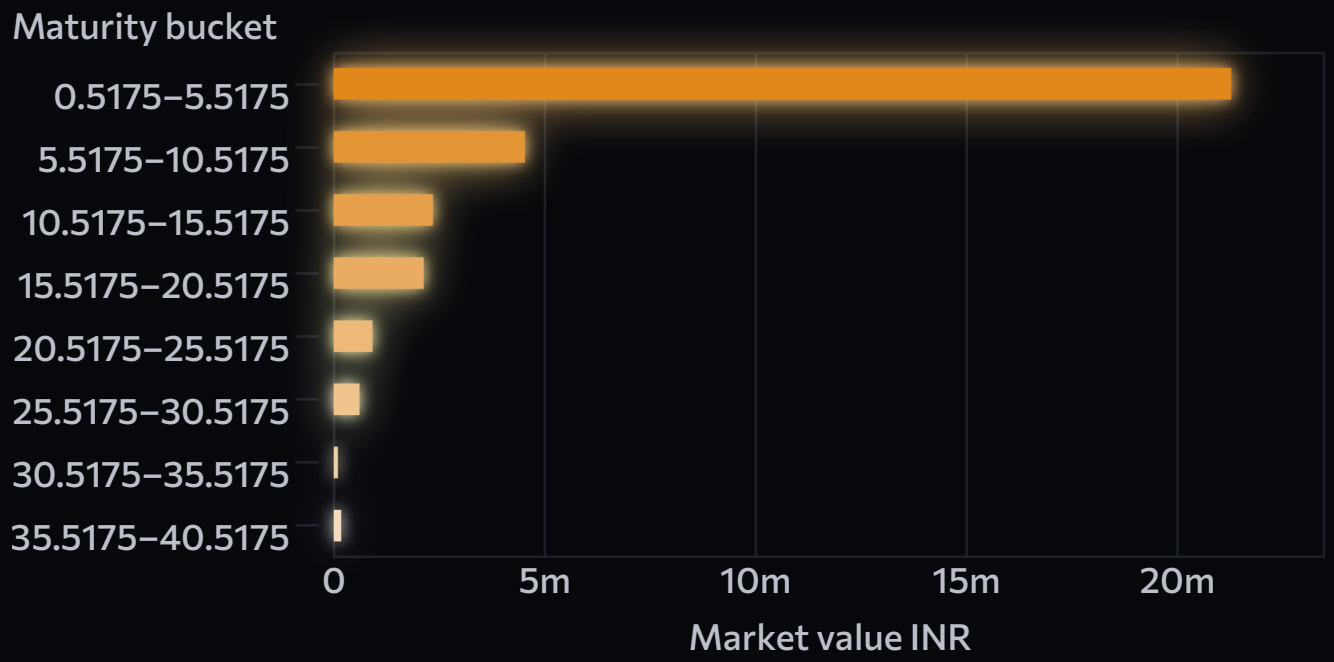
Bond count; total portfolio weight: 1.00.

₹32.06M

Market value: ₹32,059,345.97.

Government, corporate and international sovereign holdings across currencies.

FIG. 01 · MARKET VALUE BY MATURITY BUCKET



Largest maturity bucket: 0.5175–5.5175 years, at ₹21,259,033.49.

FIRST-ORDER AND SECOND-ORDER RISK

Duration is the first-order exposure; convexity adds the curve

4.2031	44.0612	₹8,948.25
Portfolio Modified Duration · years	Portfolio Convexity	Total DV01

+100 BPS PARALLEL MOVE · ANALYTICAL ESTIMATE

−4.20%	−3.98%
Duration only	Duration + convexity
First-order price sensitivity.	Curve-aware analytical estimate.

Analytical paths, not realized P&L.

CURVE RISK

Curve risk is concentrated in the long end

Latest observation: 30 October 2024 · 30Y–3M spread ≈0.91 percentage points



1.20

Peak reported KRD at 20Y

0.92

Reported KRD at 10Y

0.000829

Nelson–Siegel RMSE · R^2 ≈0.9482

SIMULATION

1,000 scenarios turn sensitivity into a distribution of outcomes

Scenarios combine parallel shift, twist and butterfly factors with duration and convexity effects.

95% TAIL

₹2.02M

Monte Carlo VaR

99% TAIL

₹2.82M

Monte Carlo VaR

95% TAIL

₹2.51M

Monte Carlo CVaR

99% TAIL

₹3.11M

Monte Carlo CVaR

VaR describes a simulated loss threshold; CVaR describes the average simulated loss beyond that threshold.

Simulated outcomes, not realized performance.

INTERPRETATION LAYERS

Machine learning and the AI Agent add interpretation layers

The AI Agent remains decision support rather than an independent decision-maker.

Random Forest	XGBoost	Neural Network
0.56	0.69	0.92
Reported R^2	Reported R^2	Reported R^2
RMSE ≈ 4.15 price points	RMSE ≈ 3.45 price points	RMSE ≈ 1.65 price points

AI Agent. Interprets duration, convexity, DV01, KRD, yield-curve scenarios, VaR and CVaR.

Predictive comparison and analytical explanation are separate jobs.

DELIVERY AND TESTING

Power BI makes the analysis explorable; Excel makes it testable

Power BI

Nine dashboard pages, more than forty DAX measures, interactive filtering, KRD, yield curve, Monte Carlo, VaR/CVaR, ML comparison and scenario analysis.

What-If: −300 bps to +300 bps in 10 bps increments.

Excel

Independently validates bond pricing, duration/convexity, portfolio aggregation, VaR and Nelson–Siegel fitting.

A second analytical path keeps the results testable.

One interface supports exploration; one workbook checks reproduction.

VALIDATION

Validation is strong where metrics reproduce, transparent where they do not

Validation area	Result
Theoretical price	Average theoretical-price difference: 0.829763424; maximum absolute difference: 2.622809435.
Portfolio metrics	Portfolio metrics reproduced: ₹32,059,345.97 market value, 4.2031 Modified Duration, 44.0612 Convexity and ₹8,948.25 DV01.
Monte Carlo versus parametric VaR	Differences were 9.91% at 95% and 11.18% at 99%, above the project's specified 5% comparison target.

Keep this as an implementation finding, not a forced pass.

FINAL REPORTING

The completed system is an analytical chain, not a trading promise

Bond data → financial calculations → portfolio analytics → simulation → machine learning → AI Agent → Power BI → Excel validation → final reporting.

Key limitations: simplified pricing, approximation error for large moves, scenario dependence, model/data dependence and the need for human review.

Thank you · Questions?